Suraj Kumar

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Education

Case Western Reserve University (CWRU) — Cleveland, OH

Master of Science (M.S.), Computer Science — *Aug 2026 - Dec 2027*

Bachelor of Science in Engineering (B.S.E.), Biomedical Engineering — *Aug 2023 - Dec 2027*

Relevant Coursework: Operating Systems (C), Intro to AI, Algorithms, Data Structures (Java), Discrete Math

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, HTML/CSS, SQL, Java, C
Frameworks & Libraries: Django, FastAPI, React.js, React Native, Next.js, PyTorch, TensorFlow, Scikit-learn
Tools: Git, AWS (ECS, VPC, RDS, S3), Docker, PostgreSQL, Linux, REST APIs, CI/CD, Gemini API, Claude Code

Software Projects

AcoustiCare | Software Engineering Lead | *AWS, Docker, Django, React Native, Expo EAS* — Nov 2025 – Present

- Led development of a voice-screening **mobile app** for ENT clinics that automates audio analysis of patient vocal health, **cutting diagnostic time from 2+ hours to seconds** per patient.
- Architected scalable AWS infrastructure (VPC, ECS, RDS PostgreSQL, S3, KMS) supporting **100+ users**, with layered security groups and TLS-terminated ALB to securely store sensitive clinical data.
- Drafted the provisional patent white paper and established a clinical data-collection pipeline with ENT physicians at UH Hospital to validate the system on real patient data.

Rival | Software Engineering Lead | *React Native, FastAPI, Gemini API, OAuth 2.0 (GitHub, Notion)* — Jan 2026

- Built a GenAI-powered 1v1 mobile game using Gemini API with structured prompt engineering to score GitHub and Notion contributions via OAuth-integrated APIs, automatically determining match winners.
- Engineered the full-stack system (matchmaking engine, real-time progress tracking, and AI verdict pipeline) to handle concurrent multiplayer matches end to end.

Work/Research Experience

AI Security Graduate Researcher — May 2026 – Present
Case Western Reserve University PEAT AI Lab — *Remote/Cleveland, OH*

- Built an unsupervised log anomaly detection framework spanning authentication logs and intrusion-detection network flows, achieving **0.98+ AUROC** in real-time threat detection across **10M+ log events**.
- Engineered TF-IDF, Word2Vec, transformer and vLLM embedders (Qwen3) paired with four outlier detection models (GMM, KDE, One-Class SVM, DeepSVDD) to convert raw log streams into actionable anomaly alerts.
- Scaled preprocessing and embedding generation on Linux HPC clusters via chunked streaming I/O and parallel SLURM array jobs, **cutting full evaluation runtime by 4x**.

Machine Learning Engineer Intern — Jun 2025 – Dec 2025
Genentech — *South San Francisco, CA*

- **Reduced the Bioassay department's experimental workload by 30%** by training machine learning models (PyTorch and Scikit-learn) to predict drug-potency experiment failures.
- Deployed ML inference endpoints to AWS EKS via GitLab CI/CD pipelines with **100%** automated test coverage.
- Optimized SQL queries across 30+ relational tables, **cutting data retrieval from 2 hours to 10 minutes**.
- Drove adoption of the analytics tool across **3 departments and 100+ scientists** through technical demos to engineers, scientists, and stakeholders.

Deep Learning Undergraduate Researcher — Sep 2024 – Nov 2025
Case Western Reserve University INVent Lab — *Remote/Cleveland, OH*

- Engineered a deep learning pipeline for glomeruli segmentation in pathology images that leverages a foundation model (MedSAM) to provide pseudo-labels for a UNet, **increasing baseline accuracy by 19%**.
- Automated extraction of 3,100 pathology slides from the KPMP database with Selenium, **cutting download time from 10 hours to 30 minutes**.
- Optimized DL algorithms on Linux HPC clusters by implementing cuDNN benchmarking and CUDA memory optimization strategies, **reducing training time by 3x**.